

Fundamentals Physics

Eleventh Edition

Halliday



Chapter 15

Oscillations

15-1 Simple Harmonic Motion (1 of 20)

Learning Objectives

15.01 Distinguish simple harmonic motion from other types of periodic motion.

15.02 For a simple harmonic oscillator, apply the relationship between position x and time t to calculate either if given a value for the other.

15.03 Relate period T , frequency f , and angular frequency ω .

15-1 Simple Harmonic Motion (2 of 20)

- 15.04** Identify (displacement) amplitude x_m , phase constant (or phase angle) ϕ , and phase $\omega t + \phi$.
- 15.05** Sketch a graph of the oscillator's position x versus time t , identifying amplitude x_m and period T .
- 15.06** From a graph of position versus time, velocity versus time, or acceleration versus time, determine the amplitude of the plot and the value of the phase constant ϕ .

15-1 Simple Harmonic Motion (3 of 20)

15.07 On a graph of position x versus time t describe the effects of changing period T , frequency f , amplitude

x_m , or phase constant ϕ .

15.08 Identify the phase constant ϕ that corresponds to the

starting time ($t = 0$) being set when a particle in SHM is at an extreme point or passing through the center point.

15-1 Simple Harmonic Motion (4 of 20)

- 15.09** Given an oscillator's position $x(t)$ as a function of time, find its velocity $v(t)$ as a function of time, identify the velocity amplitude v_m in the result, and calculate the velocity at any given time.
- 15.10** Sketch a graph of an oscillator's velocity v versus time t , identifying the velocity amplitude v_m .

15-1 Simple Harmonic Motion (5 of 20)

- 15.11** Apply the relationship between velocity amplitude v_m , angular frequency ω , and (displacement) x_m .
- 15.12** Given an oscillator's velocity $v(t)$ as a function of time, calculate its acceleration $a(t)$ as a function of time, identify the acceleration amplitude a_m in the result, and calculate the acceleration at any given time.
- 15.13** Sketch a graph of an oscillator's acceleration a versus time t , identifying the acceleration amplitude a_m .

15-1 Simple Harmonic Motion (6 of 20)

- 15.14** Identify that for a simple harmonic oscillator the acceleration a at any instant is always given by the product of a negative constant and the displacement x just then.
- 15.15** For any given instant in an oscillation, apply the relationship between acceleration a , angular frequency ω , and displacement x .
- 15.16** Given data about the position x and velocity v at one instant determine the phase $\omega t + \phi$ and phase constant ϕ .

15-1 Simple Harmonic Motion (7 of 20)

- 15.17** For a spring-block oscillator, apply the relationships between spring constant k and mass m and either period T or angular frequency ω .
- 15.18** Apply Hooke's law to relate the force F on a simple harmonic oscillator at any instant to the displacement x of the oscillator at that instant.

15-1 Simple Harmonic Motion (8 of 20)

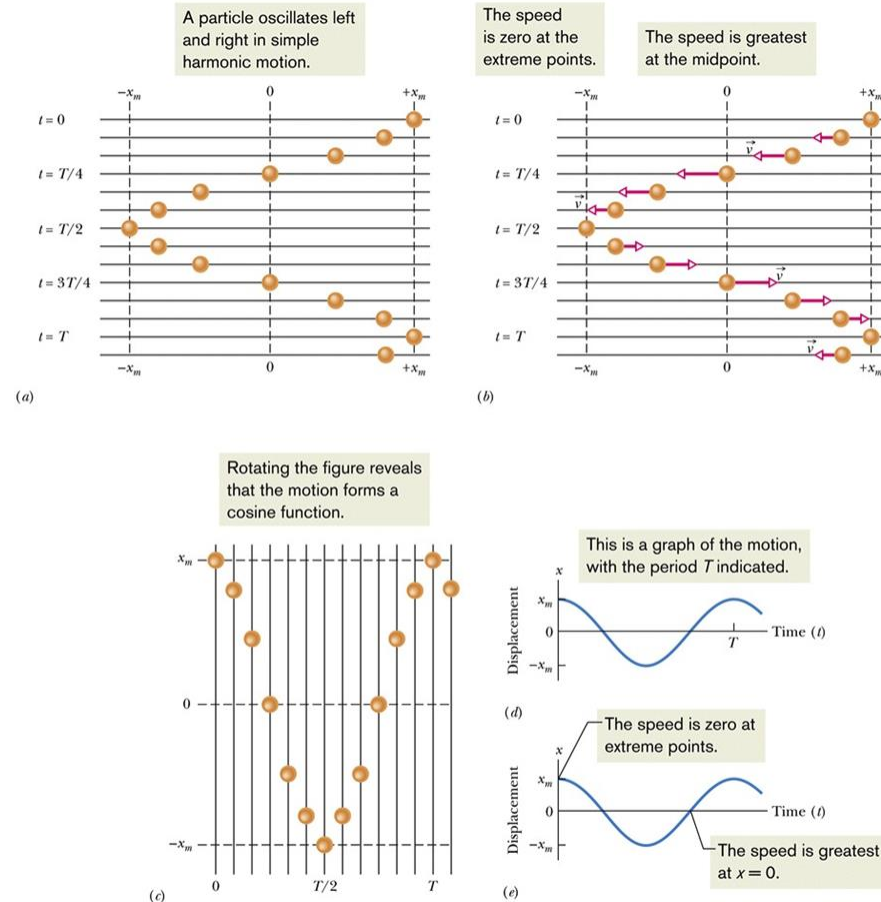
- The **frequency** of an oscillation is the number of times per second that it completes a full oscillation (cycle)
- Unit of hertz: 1 Hz = 1 oscillation per second
- The time in seconds for one full cycle is the **period**

$$T = \frac{1}{f}. \quad \text{Equation (15-2)}$$

- Any motion that repeats regularly is called periodic
- **Simple harmonic motion** is periodic motion that is a sinusoidal function of time

$$x(t) = x_m \cos(\omega t + \phi) \quad \text{Equation (15-3)}$$

15-1 Simple Harmonic Motion (9 of 20)



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Figure 15-2

15-1 Simple Harmonic Motion (10 of 20)

- The value written x_m is how far the particle moves in either direction: the **amplitude**
- The argument of the cosine is the **phase**
- The constant ϕ is called the **phase angle** or phase constant
- It adjusts for the initial conditions of motion at $t = 0$
- The **angular frequency** is written ω

15-1 Simple Harmonic Motion (11 of 20)

Displacement
at time t

Phase

$$x(t) = x_m \cos(\omega t + \phi)$$

Amplitude

Angular
frequency

Time

Phase
constant
or phase
angle

The diagram shows the equation $x(t) = x_m \cos(\omega t + \phi)$ with several labels and leader lines. 'Displacement at time t ' points to the entire equation. 'Amplitude' points to x_m . 'Angular frequency' points to ω . 'Time' points to t . 'Phase constant or phase angle' points to ϕ . 'Phase' points to the entire argument of the cosine function, $(\omega t + \phi)$.

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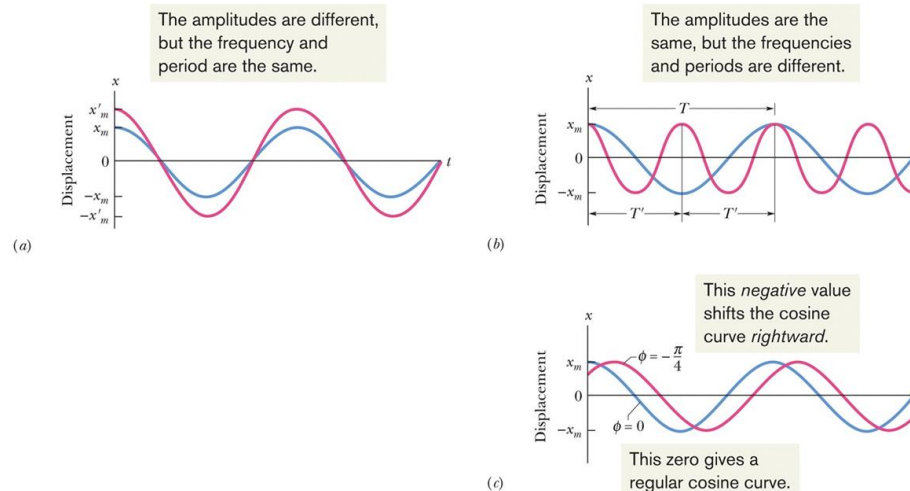
Figure 15-3

15-1 Simple Harmonic Motion (12 of 20)

- The angular frequency has the value:

$$\omega = \frac{2\pi}{T} = 2\pi f.$$

Equation (15-5)



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Figure 15-5

15-1 Simple Harmonic Motion (13 of 20)

Checkpoint 1

A particle undergoing simple harmonic oscillation of period T (like that in Fig. 15-2) is at $-x_m$ at time $t = 0$. Is it at $-x_m$, at $+x_m$, at 0, between $-x_m$ and 0, or between 0 and $+x_m$ when (a) $t = 2.00T$, (b) $t = 3.50T$, and (c) $t = 5.25T$?

Answer:

(a) at $-x_m$

(b) at x_m

(c) at 0

15-1 Simple Harmonic Motion (14 of 20)

- The velocity can be found by the time derivative of the position function:

$$v(t) = -\omega x_m \sin(\omega t + \phi) \quad \text{Equation (15-6)}$$

- The value ωx_m is the **velocity amplitude** v_m

15-1 Simple Harmonic Motion (15 of 20)

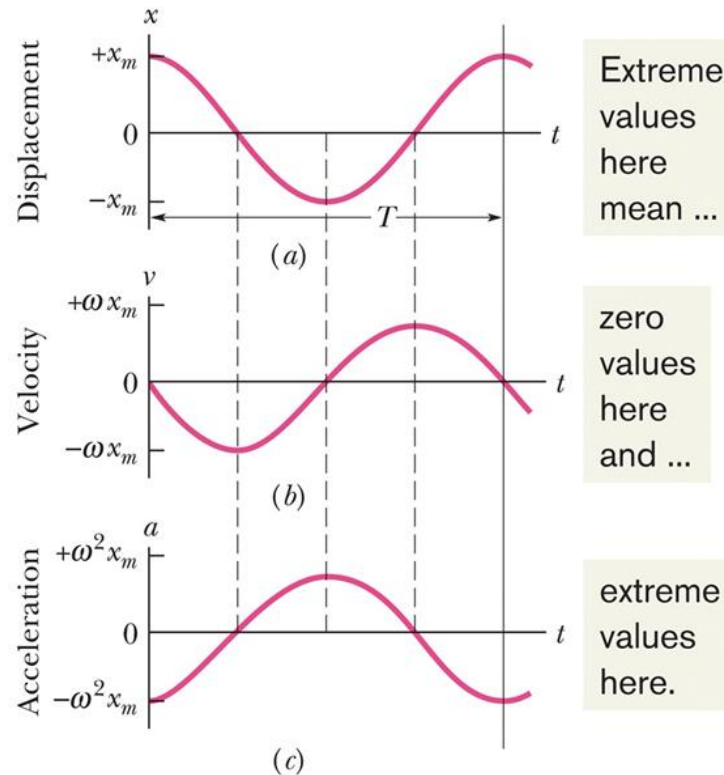
- The acceleration can be found by the time derivative of the velocity function, or 2nd derivative of position:

$$a(t) = -\omega^2 x_m \cos(\omega t + \phi) \quad \text{Equation (15-7)}$$

- The value $\omega^2 x_m$ is the **acceleration amplitude** a_m
- Acceleration related to position:

$$a(t) = -\omega^2 x(t). \quad \text{Equation (15-8)}$$

15-1 Simple Harmonic Motion (16 of 20)



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Figure 15-6

15-1 Simple Harmonic Motion (17 of 20)

In SHM, the acceleration a is proportional to the displacement x but opposite in sign, and the two quantities are related by the square of the angular frequency ω .

Checkpoint 2

Which of the following relationships between a particle's acceleration a and its position x indicates simple harmonic oscillation: (a) $a = 3x^2$, (b) $a = 5x$, (c) $a = -4x$, (d) $a = \frac{-2}{x}$?

For the SHM, what is the angular frequency (assume the unit of rad/s)?

Answer:

(c) where the angular frequency is 2

15-1 Simple Harmonic Motion (18 of 20)

- We can apply Newton's second law

$$F = ma = m(-\omega^2 x) = -(m\omega^2)x. \quad \text{Equation (15-6)}$$

- Relating this to Hooke's law we see the similarity

Simple harmonic motion is the motion of a particle when the force acting on it is proportional to the particle's displacement but in the opposite direction.

15-1 Simple Harmonic Motion (19 of 20)

- **Linear simple harmonic oscillation** (F is proportional to x to the first power) gives:

$$\omega = \sqrt{\frac{k}{m}} \quad (\text{angular frequency}). \quad \text{Equation (15-12)}$$

$$T = 2\pi\sqrt{\frac{m}{k}} \quad (\text{period}). \quad \text{Equation (15-13)}$$

15-1 Simple Harmonic Motion (20 of 20)

Checkpoint 3

Which of the following relationships between the force F on a particle and the particle's position x gives SHM: (a) $F = -5x$, (b) $F = -400x^2$, (c) $F = 10x$, (d) $F = 3x^2$?

Answer:

only (a) is simple harmonic motion (note that b is harmonic motion, but nonlinear and not SHM)

15-2 Energy in Simple Harmonic Motion (1 of 4)

Learning Objectives

- 15.19** For a spring-block oscillator, calculate the kinetic energy and elastic potential energy at any given time.
- 15.20** Apply the conservation of energy to relate the total energy of a spring-block oscillator at one instant to the total energy at another instant.
- 15.21** Sketch a graph of the kinetic energy, potential energy, and total energy of a spring-block oscillator, first as a function of time and then as a function of the oscillator's position.
- 15.22** For a spring-block oscillator, determine the block's position when the total energy is entirely kinetic energy and when it is entirely potential energy.

15-2 Energy in Simple Harmonic Motion (2 of 4)

- Write the functions for kinetic and potential energy:

$$U(t) = \frac{1}{2}kx^2 = \frac{1}{2}kx_m^2 \cos^2(\omega t + \phi). \quad \text{Equation (15-18)}$$

$$K(t) = \frac{1}{2}mv^2 = \frac{1}{2}k^2 \sin^2(\omega t + \phi). \quad \text{Equation (15-20)}$$

- Their sum is defined by:

$$E = U + K = \frac{1}{2}Kx_m^2. \quad \text{Equation (15-21)}$$

15-3 An Angular Simple Harmonic Oscillator (1 of 4)

Learning Objectives

15.23 Describe the motion of an angular simple harmonic oscillator.

15.24 For an angular simple harmonic oscillator, apply the relationship between the relationship between τ and the angular displacement θ (from equilibrium).

15-3 An Angular Simple Harmonic Oscillator (2 of 4)

- 15.25** For an angular simple harmonic oscillator, apply the relationship between the period T (or frequency f), the rotational inertia I , and the torsion constant κ .
- 15.26** For an angular simple harmonic oscillator at any instant, apply the relationship between the angular acceleration α , the angular frequency ω , and the angular displacement θ .

15-3 An Angular Simple Harmonic Oscillator (3 of 4)

- A **torsion pendulum**: elasticity from a twisting wire
- Moves in **angular simple harmonic motion**

$$\tau = -\kappa\theta.$$

Equation (15-22)

- κ is called the torsion constant
- Angular form of Hooke's law
- Replace linear variables with their angular analogs and

15-3 An Angular Simple Harmonic Oscillator (4 of 4)

- We find:

$$T = 2\pi\sqrt{\frac{I}{K}}$$

Equation (15-23)

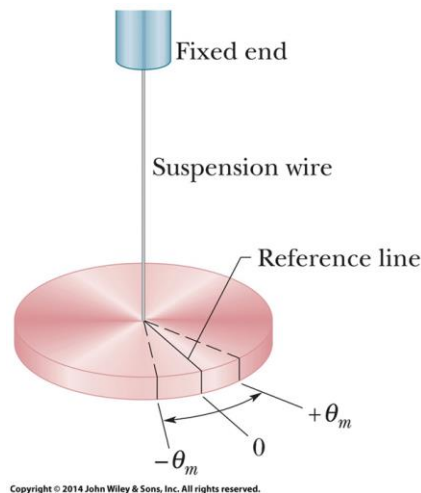


Figure 15-9

15-4 Pendulums, Circular Motion (1 of 12)

Learning Objectives

15.27 Describe the motion of an oscillating simple pendulum.

15.28 Draw a free-body diagram.

15.29-31 Distinguish between a simple and physical pendulum, and relate their variables.

15.32 Find angular frequency from torque and angular displacement or acceleration and displacement.

15-4 Pendulums, Circular Motion (2 of 12)

15.33 Distinguish angular frequency from $\frac{d\theta}{dt}$.

15.34 Determine phase and amplitude.

15.35 Describe how free-fall acceleration can be measured with a pendulum.

15.36 For a physical pendulum, find the center of the oscillation.

15.37 Relate SHM to uniform circular motion.

15-4 Pendulums, Circular Motion (3 of 12)

- A **simple pendulum**: a bob of mass m suspended from an unstretchable, massless string
- Bob feels a restoring torque:

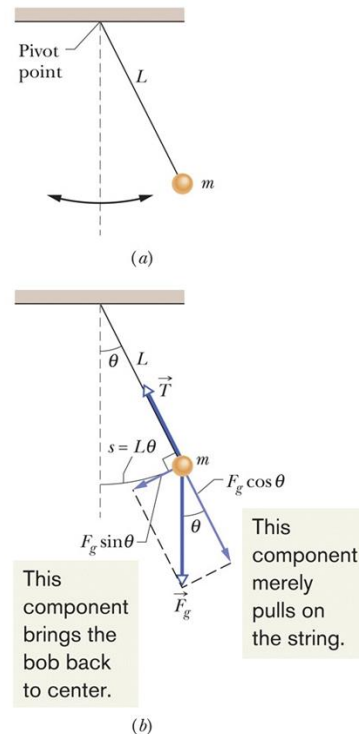
$$\tau = -L(F_g \sin \theta), \quad \text{Equation (15-24)}$$

- Relating this to moment of inertia:

$$\alpha = -\frac{mgL}{I}\theta. \quad \text{Equation (15-26)}$$

- Angular acceleration proportional to position but opposite in sign

15-4 Pendulums, Circular Motion (4 of 12)



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Figure 15-11

15-4 Pendulums, Circular Motion (5 of 12)

- **Angular amplitude** θ_m of the motion must be small
- The angular frequency is:

$$\omega = \sqrt{\frac{mgL}{I}}.$$

- The period is (for simple pendulum, $I = mL^2$):

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Equation (15-28)

15-4 Pendulums, Circular Motion (9 of 12)

Checkpoint 5

Three physical pendulums, of masses m_0 , $2m_0$, and $3m_0$, have the same shape and size and are suspended at the same point. Rank the masses according to the periods of the pendulums, greatest first.

Answer:

All the same: mass does not affect the period of a pendulum

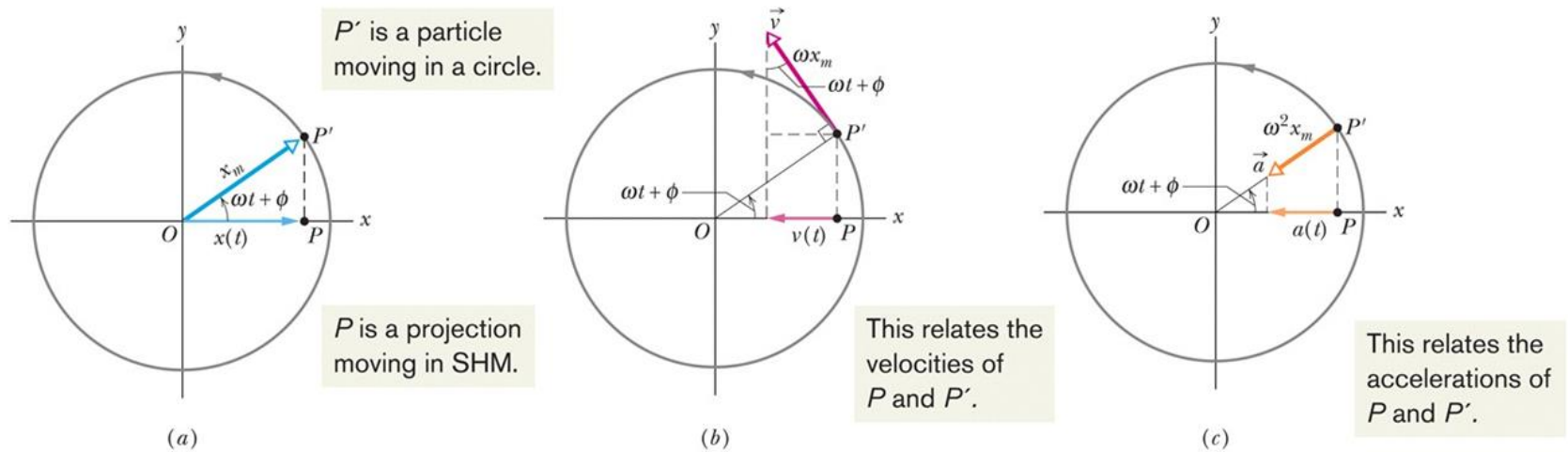
15-4 Pendulums, Circular Motion (10 of 12)

- Simple harmonic motion is circular motion viewed edge-on

Simple harmonic motion is the projection of uniform circular motion on a diameter of the circle in which the circular motion occurs.

- Figure 15-15 shows a reference particle moving in uniform circular motion
- Its angular position at any time is $\omega t + \phi$

15-4 Pendulums, Circular Motion (11 of 12)



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Figure 15-15

15-4 Pendulums, Circular Motion (12 of 12)

- Projecting its position onto x :

$$x(t) = x_m \cos(\omega t + \phi), \quad \text{Equation (15-36)}$$

- Similarly with velocity and acceleration:

$$v(t) = -\omega x_m \sin(\omega t + \phi), \quad \text{Equation (15-37)}$$

$$a(t) = -\omega^2 x_m \cos(\omega t + \phi), \quad \text{Equation (15-38)}$$

- We indeed find this projection is simple harmonic motion

15-5 Damped Simple Harmonic Motion (1 of 8)

Learning Objectives

- 15.38** Describe the motion of a damped simple harmonic oscillator and sketch a graph of the oscillator's position as a function of time.
- 15.39** For any particular time, calculate the position of a damped simple harmonic oscillator.
- 15.40** Determine the amplitude of a damped simple harmonic oscillator at any given time.

Summary (1 of 5)

Frequency

- 1 Hz = 1 cycle per second

Period

$$T = \frac{1}{f}.$$

Equation (15-2)

Simple Harmonic Motion

- Find v and a by differentiation

$$x(t) = x_m \cos(\omega t + \phi)$$

Equation (15-3)

$$\omega = \frac{2\pi}{T} = 2\pi f.$$

Equation (15-5)

Summary (2 of 5)

The Linear Oscillator

$$\omega = \sqrt{\frac{k}{m}}$$

Equation (15-12)

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Equation (15-13)

Energy

- Mechanical energy remains constant as K and U change
- $K = \frac{1}{2}mv^2, U = \frac{1}{2}Kx^2$

Summary (3 of 5)

Pendulums

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

Equation (15-23)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Equation (15-28)

$$T = 2\pi\sqrt{\frac{I}{mgh}}$$

Equation (15-29)